Te-Yi "Dexter" Kan

Los Angeles, CA (818) 309-0203 teyikan@usc.edu
dexterkan Personal Website Google Scholar

Research Summary

I am a Ph.D. candidate in ECE at USC, advised by Prof. Konstantinos Psounis. My research focuses on building efficient, privacy-preserving, and reliable ML systems for resource-constrained devices in distributed environments. Specifically, I develop privacy-preserving, low-latency pipelines for vision tasks and design reliability-aware offloading for ML workloads in vehicular networks. My recent work explores cloud–edge cooperative, hardware-aware neural architecture search for lightweight personalized models, and edge–cloud collaborative speculative decoding with confidence- and hardness-aware verification to reduce end-to-end latency, communication overhead, and cloud serving cost while maintaining output quality.

Research Interests

Privacy-Preserving ML Systems; Efficient ML Systems; Neural Architecture Search; Large-Scale Ads Systems; Scheduling; Computer Vision; Large Language Models (LLMs); Wireless Networking & Communications.

Education

University of Southern California - *Ph.D. Candidate in ECE* Aug. 2022 - May 2027 (expected)

- Dissertation Topic: Efficient and Privacy-Aware Machine Learning Systems for Resource-Constrained Devices
- Coursework: Stochastic Network Optimization, Operating Systems, Computer Vision, Artificial Intelligence, Reinforcement Learning & Large Language Models

University of California, Los Angeles - *M.S. in ECE* Sep. 2019 - Dec. 2020

- Coursework: Convex Optimization, Neural Networks and Deep Learning, Reinforcement Learning

National Taiwan University - *B.S. in EE* Sep. 2014 - Jan. 2019

- Coursework: Wireless and Mobile Networking, Wireless Communications, Network Planning and Management

Publications

[1] **Te-Yi Kan**, Konstantinos Psounis, "Neural Architecture Search for Edge Devices via Cloud-Edge Collaboration", *The International Conference on Machine Learning*, 2026, submitted. (ICML-26)

[2] **Te-Yi Kan**, Konstantinos Psounis, "Low-Latency Private ML inference for Vision Tasks in Distributed Environments", *IEEE Transactions on Dependable and Secure Computing*, 2026, submitted. (TDSC-26)

[3] **Te-Yi Kan**, Konstantinos Psounis, "Online distributed offloading of time-sensitive vehicular tasks in edge-cloud systems", *IEEE Transactions on Vehicular Technology*, 2025. (TVT-25) |

[4] **Te-Yi Kan**, Ronald Y. Chang, Feng-Tsun Chien, "Hybrid Intelligent Reflecting Surface and Classical Relay Assisted Multiuser MISO Systems", *IEEE Transactions on Vehicular Technology*, 2023. (TVT-23) |

[5] Tz-Wei Mo, Ronald Y. Chang, **Te-Yi Kan**, "DeepMCTS: Deep Reinforcement Learning Assisted Monte Carlo Tree Search for MIMO Detection", *2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring)*, 2022. (VTC-22) |

[6] **Te-Yi Kan**, Ronald Y. Chang, Feng-Tsun Chien, "Intelligent Reflecting Surfaces and Classical Relays: Coexistence and Co-Design", *2021 IEEE GlobeCom Workshops (GC Wkshps)*, 2021.(GC-21). |

[7] **Te-Yi Kan**, Yao Chiang, Hung-Yu Wei, "QoS-aware Mobile Edge Computing System: Multi-server Multi-user Scenario", *2018 IEEE GlobeCom Workshops (GC Wkshps)*, 2018. (GC-18)

[8] **Te-Yi Kan**, Yao Chiang, Hung-Yu Wei, "QoS-aware Fog Computing System: Load Distribution and Task Offloading", *The 15th IEEE Vehicular Technology Society Asia Pacific Wireless Communications Symposium (VTS APWCS)*, 2018. (APWCS-2018) |

[9] **Te-Yi Kan**, Yao Chiang, Hung-Yu Wei, "Task Offloading and Resource Allocation in Mobile-Edge Computing System", *2018 27th Wireless and Optical Communication Conference (WOCC)*, 2018. (WOCC-18) |

Research Projects

Collaborative Speculative Decoding Across Edge and Cloud: Confidence- and Hardness-Aware Drafting and Verification Feb. 2026 - Present

- Built a confidence- and hardness-aware edge–cloud speculative decoding pipeline that dynamically decides whether draft tokens should be uploaded for cloud verification (token-level selective verification).
- Reduced end-to-end latency by avoiding unnecessary verification, cutting both cloud compute time and edge-cloud communication overhead by sending only a subset of "hard/uncertain" tokens.
- Improved cloud resource efficiency and serving cost by lowering verifier load through adaptive gating, while maintaining output quality via targeted verification of high-risk tokens.

Efficient Hardware-aware Neural Architecture Search for Mobile Devices [*ICML-26*] Jun. 2025 - Jan. 2026

- Designed a two-stage NAS framework combining cloud-based search with edge-based fine-tuning, enabling high-performance personalized models under strict latency and memory constraints.
- Designed a lightweight, restartable on-device search (sequential beam search) with caching, stage-aware evaluation, and intermediate feature reuse, enabling rapid re-optimization under time-varying edge resources.
- Validated the framework on vision (MobileNet-style) and NLP (GPT-style) search spaces, achieving $2.7\times$–$3.8\times$ lower search time while maintaining competitive utility.

***Privacy-Preserving ML Systems* [*TDSC-26*]** Jun. 2024 - Jun 2025

- Built a privacy-preserving inference pipeline with sensitive object detection, task scheduling, and obfuscation modules for vision tasks.
- Leveraged distributed remote servers to balance utility and privacy.
- Achieved $\sim$30% higher task utility at 25 fps without compromising user privacy.

Online Distributed Offloading Mechanism in Vehicular Networks [*TVT-25*] Feb. 2023 - May 2024

- Developed TODORA, a threshold-based online distributed offloading and resource allocation mechanism for real-time vehicular ML tasks.
- Reduced end-to-end task duration (execution delay + communication overhead) by optimizing offloading and allocation under dynamic edge–cloud environments.
- Demonstrated superior performance over state-of-the-art baselines via extensive simulations.

Intelligent Reflecting Surface (IRS) [*TVT-23, GC-21*] Jan. 2021 - May 2022

- Proposed a multiuser downlink MISO system assisted by a full-duplex IRS and half-duplex DF relay.
- Designed an alternating-optimization algorithm to jointly optimize active BS/relay beamforming and IRS passive beamforming for maximum sum-rate.
- Demonstrated significant throughput gains and analyzed tradeoffs in joint beamforming design.

Multiple-input multiple-output (MIMO) Detection [*VTC-22*] Oct. 2020 - Apr. 2021

- Devised a deep reinforcement learning enhanced Monte Carlo Tree Search (MCTS) framework for MIMO detection.
- Demonstrated notable performance and complexity improvements over conventional MCTS across diverse channel conditions.

Multi-access Edge Computing (MEC) Systems [*GC-18, APWCS-18, WOCC-18*] Sep. 2017 - Aug. 2018

- Developed QoS-aware task offloading strategies for MEC systems in single- and multi-server settings.
- Proposed a two-stage algorithm optimizing task offloading, resource allocation, and load distribution.
- Validated superior latency reduction and QoS performance against multiple benchmarks.

Professional Activities

Reviewer, IEEE & ACM Sep. 2020 - Present

- Journals: *IEEE Access*, *IEEE Transactions on Vehicular Technology*, *IEEE Transactions on Green Communications and Networking*, *IEEE Internet of Things Journal*
- Conferences: *IEEE Wireless Communications and Networking Conference*, *IEEE International Conference on Communications*, *IEEE International Symposium on Personal, Indoor and Mobile Radio Communications*, and *The International Conference on Machine Learning (ICML)*.

Teaching Experience

USC EE450: Introduction to Computer Networks Spring, Summer, and Fall 2023
Instructors: Dr. Ali Zahid, Dr. Sharin Nazarin, and Dr. Corey Baker

USC EE503: Probability for Electrical and Computer Engineers Fall 2022
Instructors: Dr. Konstantinos Psounis

Technical Skills

Programming Languages:	Python, C/C++, Java, Bash, Matlab, R
ML & AI:	PyTorch, TensorFlow, Scikit-learn, Keras, Transformers
Infrastructure & Deployment:	OpenStack, Docker, Kubernetes, Pandas

Work Experience

***Machine Learning Engineer Intern*, Meta** May 2026 - Aug. 2026
- Improved large-scale ads attribution models through architecture, feature, and objective optimization, executing 200+ A/B experiments that identified promotable MLX candidates.
- Built an AI-agent-driven experimentation pipeline that accelerates the model-improvement loop, reducing experiment reruns from days to hours.
- Partnered cross-functionally with upstream teams to build training-data pipelines that onboarded new labels and event-based sequence features, expanding modeling signals for experimentation.

***Researcher*, Academia Sinica-MediaTek Collaborative Project** Oct. 2021 - May 2022
Taipei, Taiwan
- Proposed and collaborated with MediaTek research team on the project, "AI-Enabled Hybrid Reconfigurable Intelligent Surface (RIS) and Relay Systems for 6G Communications."
- Designed an AI-enabled hybrid RIS/relay system to improve energy efficiency and system sum-rate.
- Analyzed system performance and tradeoffs in mmWave and THz frequency bands.

***Summer Intern*, Foxconn Technology Group** Jun. 2017 - Aug. 2017
Taipei, Taiwan
- Implemented a commercial MEC infrastructure to enable lifecycle management of MEC services.
- Contributed to the development of an MEC-powered face recognition system for access control and attendance management, using OpenStack for deployment.

Presentations

Poster
- "Low-Latency Private ML Inference for Vision Tasks in a Distributed Environment," The ECE 14th Annual Research Festival, Los Angeles, CA, Nov. 8, 2024
- "Joint Task Assignment and Resource Allocation Mechanism for In-vehicle Augmented Reality," The ECE 13th Annual Research Festival, Los Angeles, CA, Oct. 27, 2023
- "QoS-aware Mobile Edge Computing System: Multi-server Multi-user Scenario," IEEE Globecom Workshops (GC Wkshps), Abu Dhabi, UAE, Dec. 09, 2018

Presentation
- "Intelligent Reflecting Surfaces and Classical Relays: Coexistence and Co-Design," IEEE Globecom Workshops (GC Wkshps), Madrid, Spain, Dec. 07, 2021
- "QoS-aware Fog Computing System: Load Distribution and Task Offloading," IEEE Vehicular Technology Society Asia Pacific Wireless Communications Symposium (IEEE VTS APWCS), Taiwan, Aug. 23, 2018
- "Task Offloading and Resource Allocation in Mobile-Edge Computing System," Wireless and Optical Communication Conference (WOCC), Taiwan, May 01, 2018

Selective Projects

Schematic-to-Netlist Translation using RL-Enhanced VLM Pipeline Fall 2025
- Developed an RL-enhanced VLM pipeline to translate circuit schematic diagrams into SPICE-compatible netlists.

- Trained Qwen with LoRA for schematic understanding, then used GRPO with syntax and functional correctness as reward signals to iteratively refine translation quality.

Operating System Implementation Spring 2024

- Completed Brown University's kernel assignment project to build a simple operating system called Weenix.
- Implemented essential elements for an operating systems, including but not limited to process, thread, scheduler, virtual file system, and virtual memory.

Drift-Plus-Penalty (DPP)-based Task Offloading Mechanism for Vehicular Networks Fall 2023

- Devised a DPP-based algorithm to effectively schedule resource-intensive tasks on edge and cloud servers.
- Verified the proposed algorithm can achieve comparable performance in low complexity as compared to the optimal solution.

Reinforcement Learning (RL)-Based Mechanism for Loss Reduction during the COVID-19 Outbreak Spring2020

- Developed RL-based algorithms to provide appropriate business strategies for the food and beverage industry.
- Simulated the proposed RL-based algorithms to verify the superiority of our proposed methods in comparison to baselines.

Generative Adversarial Imitation Learning (GAIL) Spring 2020

- Implemented a scalable imitation learning algorithm in Python for high-dimensional environments.
- Achieved superior performance over Behavior Cloning on OpenAI Gym tasks (Pendulum, Cartpole).

Image Generation Methods for Cataract Surgery Winter 2020

- Applied U-Net and U-Net+WGAN frameworks to segment cataract surgery images of eye structures.
- Benchmarked models using pixel accuracy, mean IoU, and F1 score, with comparisons against self-labeled data.

IEEE 802.15.4 Design Challenge Fall 2017

- Designed duty cycle–adaptive Machine-to-Machine (M2M) communication protocols for wireless networks.
- Implemented protocols on FCM2401 modules using AutoNet SDK.

Intra System Handoff in Heterogeneous Wireless Networks NTU

- Simulated the performance of mobile vertical and horizontal handoff among 3G, 4G, and WiFi in the Taipei Metro System.

Text Recognizing Voice Recorder NTU

- Established a novel design with knowledge in digital signal processing and circuit design of a voice recorder on FPGA able to automatically recognize text commands and perform corresponding functions.

Functionally Reduced And-Inverter Graph (FRAIG) Fall 2016

- Built a C-based FRAIG circuit representation with data structures for detecting functionally equivalent pairs.
- Integrated hashing, Boolean simulations, and SAT solvers to identify equivalences in digital circuits.

Honors and membership

- Viterbi Graduate Fellowship, Fall 2022.
- Undergraduate Overseas Research Presentation Grant, Fall 2018.
- Student member, IEEE and IEEE Communications Society, since 2018.